

INT6068 Neural Networks and Deep Learning

Group project

Objectives:

1. To show understanding of deep learning and neural networks;
2. To realize related technologies with practical applications.

Topics:

You may choose any one from the following topics. Consult the lecturer before you develop your project.

1. Reproduce the algorithms published in one or two technical papers related to machine learning. You may find those papers from CVPR, ICCV, ECCV, TPAMI, TIP or other resources.
2. Develop your own algorithms for an application problem.
3. Other topics.

Presentation:

1. The presentation should include: technical details of the algorithms, dataset details, accuracy, sample results, speed. Better to include ablation study, hyperparameter optimization or more.
2. Each group has up to 20 minutes (18 minutes for presentation, 2 minutes for Q&A). Each group member should present their contributions.
3. The powerpoint slides of the presentation (*.ppt) and the program (*.py) should be submitted **on or before the due date**, which is shown on **Moodle**.

Evaluation Criterion:

Item	Grading criteria	Weight
1. Presentation	technical soundness, and presentation skills	60%
2. Program	correctness, difficulty, readability, and robustness	30%
3. Powerpoint slides	technical soundness, presentation skills	10%

Contribution of group members

Contribution of each group member should be elaborated in the Powerpoint slides.